

Part 1 of 3

Towards an Extra Layer of Security: Two-Factor Authentication



Hari Om Prakash is working as a scientist at Systems Engineering Group, ADRIIN, Department of Space

The buzz in the web market is two-factor authentication (also known as TFA, 2FA, two-step verification, multi-factor authentication). Many users hold multiple accounts on multiple websites, and most of them use the same password for multiple accounts. If their password is hacked on one website, their accounts on other websites get vulnerable too.

Passwords are not secure enough—if someone gets your password, they can easily access your account and can do anything with your account like deleting your data, sharing your sensitive data with the public, hijacking, etc.

By using two-factor authentication, it becomes more difficult for hackers to impersonate you and take over your account. With an increase in cyber attacks or exploitation of the first level, there is need for a second level of authentication that adds more security to services or contents like e-mail, bank accounts, social accounts, etc.

What is two-factor authentication

Authentication is concerned with assuring an authentic communication. Till now, you may be aware of one-factor authentication only, where you are asked for a user login name and password (for example: Gmail login in Fig. 1 and SBI online net banking login in Fig. 2).

The screenshot shows the Gmail login interface. At the top is a grey circle icon representing a user profile. Below it are two input fields: 'Email' and 'Password'. A blue 'Sign in' button is positioned below the password field. At the bottom left, there is a checkbox labeled 'Stay signed in', and at the bottom right, there is a link that says 'Need help?'.

Fig. 1: Gmail login (Credit: Google/Gmail)

The screenshot shows the State Bank of India (SBI) online net banking login page. At the top is the SBI logo and the text 'भारतीय स्टेट बैंक State Bank of India The Banker to Every Indian'. Below this is a navigation bar with links: 'Home', 'Products & Services', and 'How Do I'. The main heading is 'Login' with a sub-heading 'Welcome to Personal Banking'. The login form includes the text 'To access your accounts... Login to OnlineSBI' and a warning '(CARE: Username and password are case sensitive.)'. There are input fields for 'Username *' and 'Password *'. Below these is a checkbox 'Enable Virtual Keyboard'. At the bottom are three buttons: 'Login', 'Reset', and 'Forgot Login Password'.

Fig. 2: SBI login (Credit: State Bank of India)